

Longitudinal Vehicle Performance Simulation and Gear-Shift optimisation

A First-Principles Model of Engine, Drivetrain and Straight Line Vehicle Dynamics (in Python)

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1 Abstract

This study develops a simplified first-principles longitudinal vehicle performance model in Python, investigating the influence of gear-shift strategy on acceleration and high speed performance. The model combines a synthetic engine torque curve, drivetrain, gearing, tractive force and resistive forces like aerodynamic drag and rolling resistance within a time domain simulation. Gear shift points were determined from the cross over in tractive force between adjacent gears and compared to a baseline strategy in which gear shifting occurred at the 10,000 RPM engine speed limit. The optimised strategy reduced the acceleration time for the 0-200 *km/h* event from 18.90s to 17.12s, approximately 9.4%, whilst producing a higher simulated equilibrium speed. The study demonstrates how a simplified model can be used to evaluate and optimise drivetrain control decisions, whilst highlighting the limitations that come with synthetic engine and vehicle parameters.

2 Introduction

Vehicle acceleration performance on the interaction between the engine, drivetrain, gearing and resistive forces acting on the vehicle. The engine converts different amounts of torque and engine speed through a selected gear ratio. As a result the selected gear has a direct influence on the tractive force available and therefore impacts the longitudinal acceleration.

The simplified approach to shifting in this study is shifting when the engine reaches its maximum operating speed i.e. redline. This, however, does not provide the best acceleration to the vehicle. At high engine speeds, engine torque may decrease and the shifting into the next gear will result in a greater tractive force despite the reduction in the gear ratio. This creates an engineering

opportunity, to determine the shift points based on the tractive force cross over between adjacent gears instead of relying solely on engine speed.

This study develops a simplified first principles longitudinal model in Python to investigate this behaviour. A synthetic torque curve for the engine is combined with a six speed drive train model, resistive forces and a time domain simulation. Shifting points are determined by comparing the tractive force available in adjacent gears and are then used to develop an optimised shift strategy through different events: 0-100 *km/h*, 0-200 *km/h* and finally, a high speed event in which the target speed was set to 350 *km/h*

3 Engine Model

3.1 The Objective

The first stage of the model was to develop a simple representation of the engine performance across the operating speed range. The engine model provides the torque at a given engine speed which will be used by the drivetrain model to determine wheel torque and tractive force.

Engine torque was represented as a function of the engine speed, shown below:

$$T_e = f(N) \tag{1}$$

Where T_e is the engine torque in Nm and N is engine speed in Revolutions per minute (RPM)

To make modelling easier, synthetic data (not real) was made and used rather than data specific from a production vehicle. This was done to develop the modelling methodology and test it independent of a vehicle's data.

3.2 Torque Data

The data shown below is the synthetic data used for this project. Figure 1 below the data displays the tabulated data in a graph and shows how the torque initially increases with engine speed and reaches a maximum of 320 Nm at 5000 RPM and then decreases subsequently as it reaches 10000 RPM (the upper bound of the operating range)

Engine Speed (RPM)	Torque (Nm)
1000	150
2000	200
3000	250
4000	290
5000	320
6000	315
7000	295
8000	260
9000	220
10000	150

Table 1: Synthetic data used for the initial model

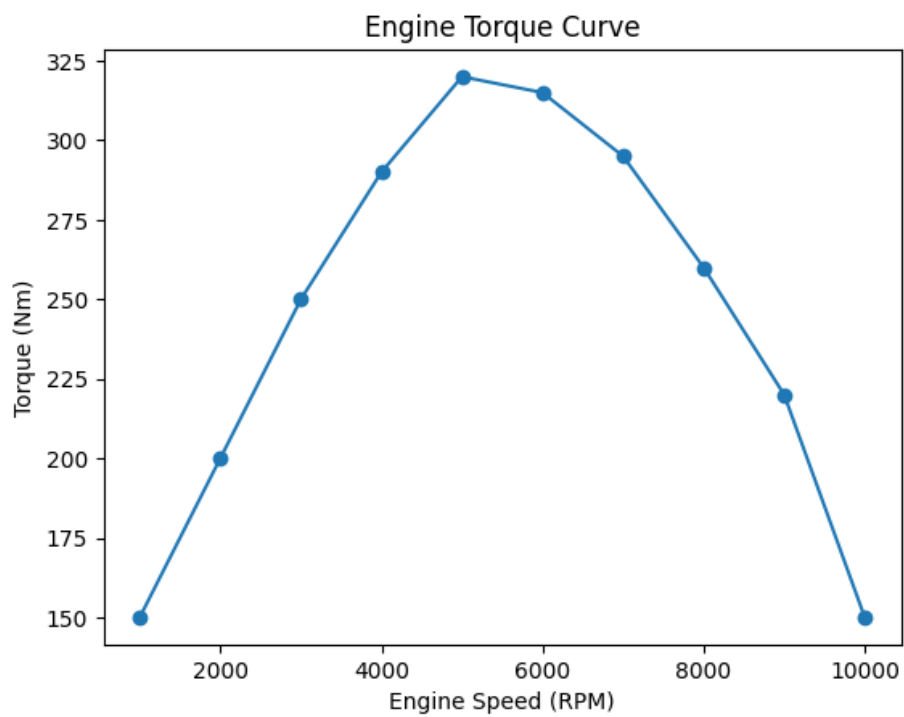


Figure 1: Engine Torque Curve

3.3 Engine Power

The relationship relating torque and angular velocity is given by the equation

$$P = T\omega \quad (2)$$

Where P is Power in Watts, T is Torque in Nm and ω is angular velocity in radians per second (rad/s). The engine speed data was specified in RPM and therefore was converted to angular velocity with the following equation:

$$\omega = \frac{2\pi N}{60} \quad (3)$$

Power was calculated from the interpolated torque, T and its corresponding angular velocity ω

The RPM corresponding to the maximum torque is different to RPM corresponding to the maximum power. Torque begins to decrease after 5000 RPM whereas engine speed continues to increase. As shown in equation 2, Power depends on the product of both torque and angular velocity and the increase in angular velocity outweighs (initially) the decrease in torque, allowing engine power to continue increasing despite torque going down.

The model predicts the maximum power at around 8000 RPM with the maximum torque being around 5000 RPM

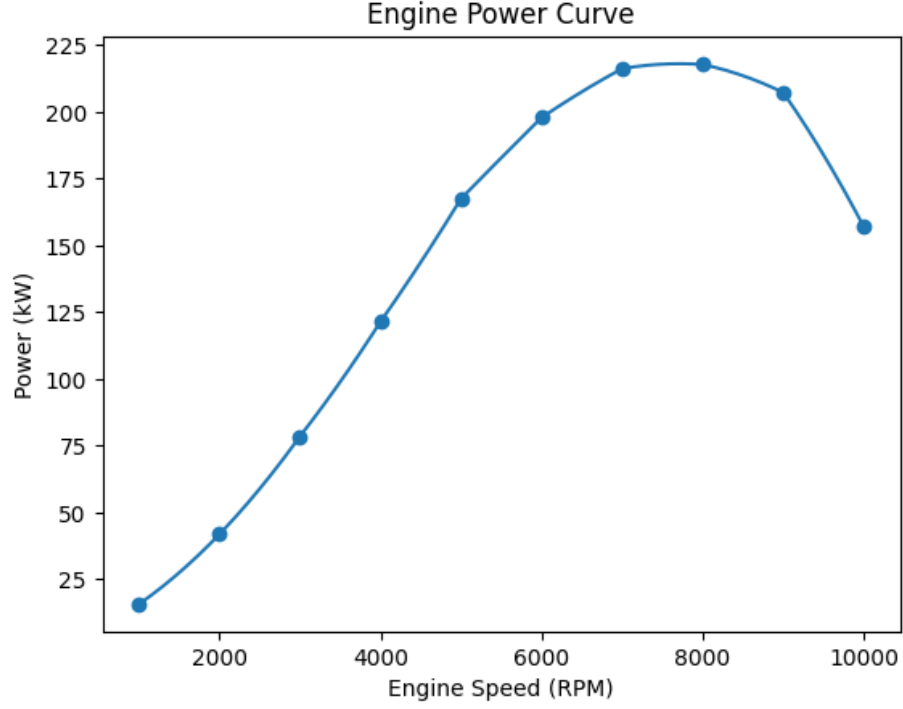


Figure 2: Engine Power Curve

3.4 Torque Interpolation

The original RPM-Torque data contained values at 1000 RPM intervals. However its important to note that the during a dynamic simulation the engine will not operate at intervals of 1000 RPM and the drivetrain model could require torque at an RPM of 4510 instead of exactly 4000 or 5000 RPM. As a result linear interpolation was used to estimate the torque between two points.

$$Y(X) = Y_1 + \frac{X - X_1}{X_2 - X_1}(Y_2 - Y_1) \quad (4)$$

For example, the between 5000 and 6000 RPM where the respective torques are 320 Nm and 315 Nm, the linear interpolation results in a torque of 317.5 Nm at 5500 RPM. However, in this project this was not carried out by hand but rather done by using NumPy's linear interpolation function `np.interp()`

To make analysis and visualisation significantly easier, the model was evaluated at 50 RPM intervals. Note: That increasing the number of points does not increase the physical accuracy of the underlying model. To improve the accuracy, higher quality data would be required.

4 Drivetrain and Gearing

4.1 Drivetrain Model

The model developed in the previous section provided the torque available at a given engine speed. However, the engine torque doesn't transmit to the wheels directly. The torque passes through the gearbox, driveshaft and then final drive, altering the relationship between torque and angular velocity before the power is delivered to the wheel.

The synthetic data used by the drivetrain model is presented below. It shows a six-speed gearbox with each gear having a fixed gear ratio.

Gear (Number)	Gear Ratio
1	3.50
2	2.20
3	1.55
4	1.20
5	1.00
6	0.80

Table 2: Synthetic data used for the drivetrain model

A hard coded final drive ratio of 4.0 along with a drivetrain efficiency of 0.90 was assumed. Additionally, the wheel's rolling radius was set as 0.33 m. As mentioned previously, the data is synthetic for the purpose of model development and evaluation.

The wheel torque calculation was carried out using the formula below:

$$T_w = T_e G_i G_f \eta \quad (5)$$

where T_w is equal to the wheel torque, T_e is the engine torque, G_i is the gear ratio (of the gear in use), G_f is the final drive ratio and η is the drivetrain efficiency.

The gearbox is used to provide torque multiplication at the expense of the rotational velocities, which is why the higher ratios are reserved for lower gear number to allow for greater wheel torque which allows acceleration whereas the higher gears allow for higher rotational velocities but lower acceleration.

4.2 Wheel Speed and Vehicle Speed

The relationship between wheel speed and vehicle speed is an important relationship to evaluate. The equation below was used to determine the angular velocity of the wheel ω_w using the selected gear ratio G_i and final drive ratio G_f

$$\omega_w = \frac{\omega_e}{G_i G_f} \quad (6)$$

The vehicle's velocity v was then evaluated using the effective rolling radius of the wheel r_w and the angular velocity ω_w

$$v = \omega_w r_w \quad (7)$$

An ideal gearbox conserves mechanical power while trading angular velocity for torque. Lower gears use larger ratios to increase wheel torque while reducing the angular velocity of the wheel for a given engine speed. Additionally, drivetrain losses are represented separately by the efficiency η . For example, at 5000 RPM and in gear 1 with a gear ratio of 3.50, the model evaluated a wheel torque of approximately 4000 Nm whereas at gear 6, with a gear ratio of 0.80, the wheel torque was approximately 920 Nm. The vehicle speeds at those wheel torques was evaluated at approximately 44 *km/h* and 194 *km/h*.

Note: These speeds correspond to a given engine speed and gear ratio. They do not imply that the vehicle is capable of reaching each operating point. Aerodynamics drag, rolling resistance and the tractive force available determine whether a vehicle can reach the corresponding speed.

4.3 Tractive Force and how it varies across the Gear range

Wheel torque has to be converted to the longitudinal force before the vehicles motion can be modelled. The force generated at the contact points of the tyre and road was evaluated using the equation below:

$$F_t = \frac{T_w}{r_w} \quad (8)$$

Where F_t is the tractive force, T_w is the wheel torque and the effective rolling radius is r_w

Tractive force is the longitudinal force transmitted through the contact point between the road and the tyre, which acts to push the vehicle forwards.

The complete engine torque curve was evaluated across the 1000-10,000 RPM range for each of the gears. For each engine speed, the wheel torque, tractive force and vehicle speed were calculated, which then allowed visual representation of the tractive force-vehicle speed curve (shown below):

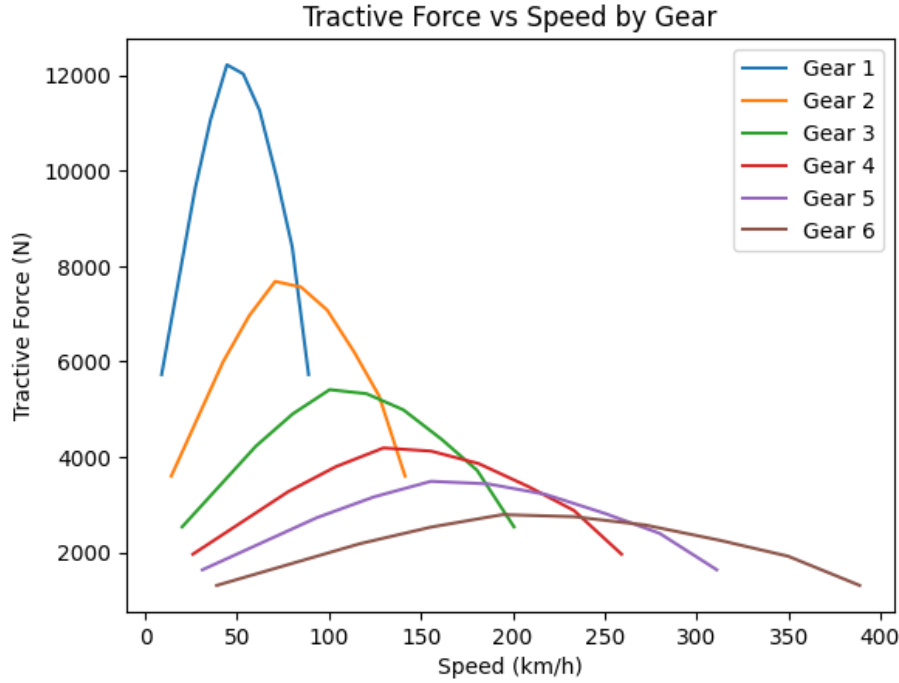


Figure 3: Tractive Force vs Vehicle Speed

The results show how gearing can effect the performance of a vehicle. First gear produces the largest tractive force but has the smallest vehicle speed range out of the six gears whilst the higher gears provide lower tractive force (as the gear number increases) but have significantly larger vehicle speeds and larger ranges as well. It goes without saying that although the graph above seems to indicate that gear six can technically work from 1000 RPM to 10,000 RPM in actuality this would not work due to external conditions such as engine stall.

Continuing with the graph above, each curve increases initially and the decreases as the shape is dependent on the engine torque curve. As mentioned above, the final-drive ration, drive train efficiency and the wheel radius are constant values, the primary driver behind the changing tractive curve and therefore the shape of the curves is the engine torque changing with RPM.

The torque curve also shows how adjacent curves intersect. At high enough engine speeds engine torque begins to decrease significantly, therefore shifting into the next gear will cause the engine to have lower RPM but higher engine torque will be available and consequently higher wheel torque will also be available.

The crossover points provide optimal points to shift and will be determined to then construct the gear-shift strategy.

5 Determining Gear-Shift Point

The graph for the tractive force for each gear show that adjacent gears overlap across portions of vehicle speed ranges. Within the overlapping regions there a point where the current gear is worse then next gear at a given vehicle speed. The method to determine this point utilised tractive forces developed by each gear.

$$\Delta F(v) = F_{t,i}(v) - F_{t,i+1}(v) \quad (9)$$

Where $F_{t,i}$ is the tractive force of the current gear and $F_{t,i+1}$ is the tractive force of the next gear.

From the equation above it can be clearly seen that when $\Delta F > 0$ the current gear provides greater tractive force then the next gear at that vehicle speed. However, when $\Delta F < 0$ the subsequent gear i.e. gear $i + 1$ provides greater tractive force. As a result, the ideal gear change i.e. shift point occurs when $\Delta F = 0$

Since each gear produces a different set of vehicle speed values across the engine operating range, the two arrays containing tractive forces for gear i and gear $i + 1$ cannot be compared on an element by element basis. For each pair of gears the common vehicle speed was identified and a shared speed grid was made. Linear interpolation was then used to evaluate the tractive force of both gears at identical vehicle speeds. The difference between the two interpolated curves was then calculated across the common range and a change in sign from $+\Delta F$ to $-\Delta F$ was used to identify the point at which the next gear began to provide greater tractive force.

The vehicle speed was then converted back into the engine speed of the current gear by rearranging the formula below:

$$v = w_\omega r_\omega \quad (10)$$

By obtaining the wheel's angular velocity, w_ω the engine angular velocity and then consequently the RPM can be determined using the formulae below:

$$w_e = w_\omega G_i G_f \quad (11)$$

$$N_e = w_e \frac{60}{2\pi} \quad (12)$$

Where N_e is the engine RPM.

The table below shows the gear changes at their respective vehicle speeds and the RPM just before shifting.

Gear Change	Shift Speed (km/h)	Approximate Engine RPM before shifting
1→2	82.7	9300
2→3	127.9	9050
3→4	174.1	8680
4→5	215.8	8330
5→6	265.38	8530

Table 3: Gear Change, Speed and Pre-Shift RPM

The results demonstrated that the optimum shifting point is not equal to the 10,000 RPM operating speed of the engine. As engine torque decreases at high rotational speeds, going up to the next gear can return the engine to a higher torque region where RPM will be lower.

Each up-shift produces a different reduction in engine speed, as a result it places the engine at a different position on its torque curve.

It is important to note that a limitation to this model and project overall is that it assumes that gear change is instantaneous and that there is no interruption or loss of tractive force during the gear change process. Further limitations are discussed at the end of this report.

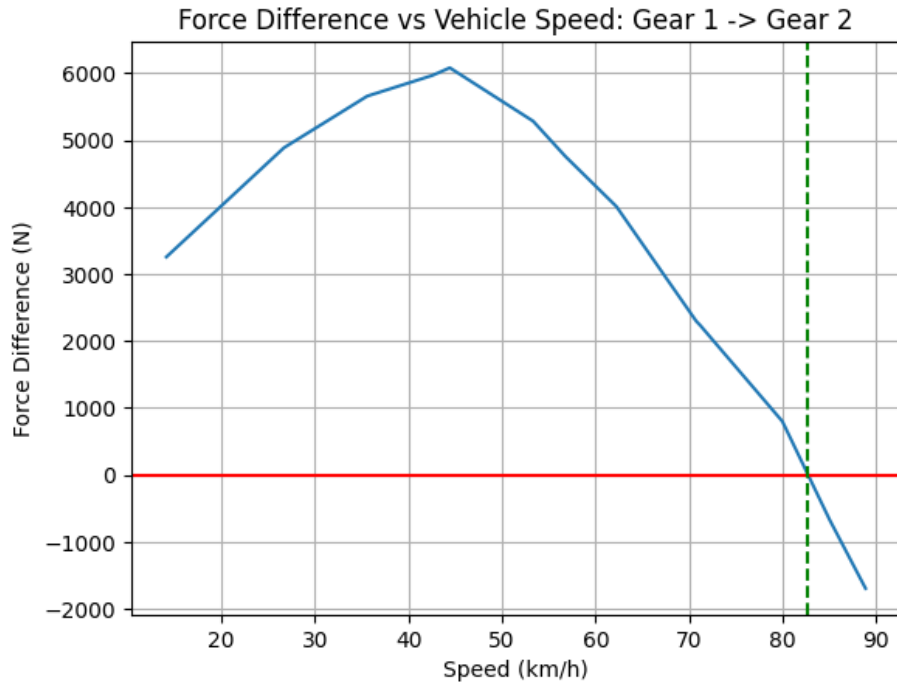


Figure 4: ΔF vs Speed (Gear 1→2)

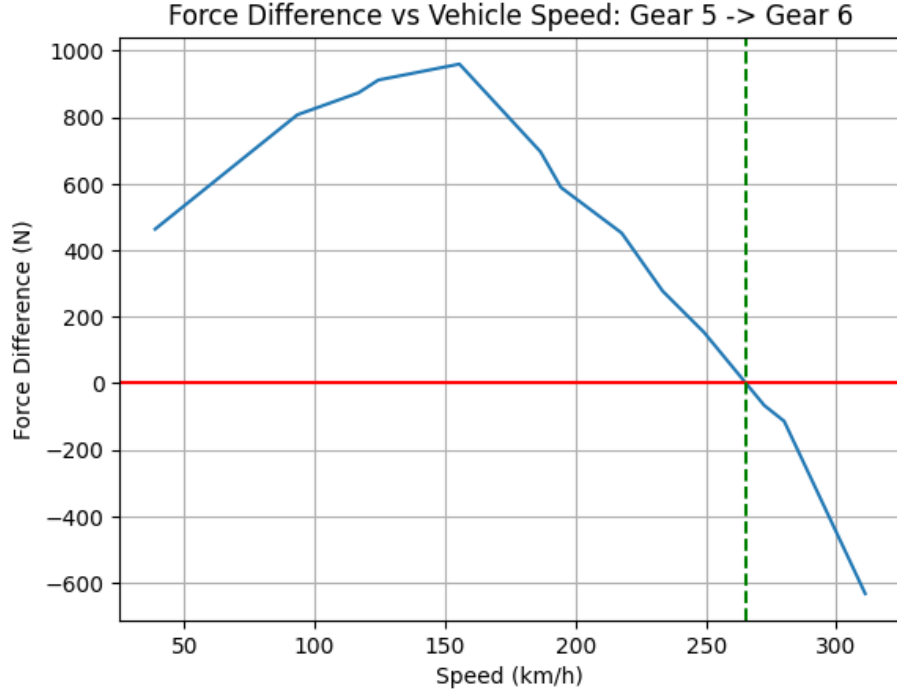


Figure 5: ΔF vs Speed (Gear 5→6)

Figures 4 and 5 demonstrate the change in tractive forces at a given speed and shows how it varies with speed. The intersection of curve with the red line i.e. when $\Delta F = 0$ as marked by the green line is the point where the shift change occurs and at that point the next gear provides greater tractive force compared to the current gear.

6 Resistive Forces

6.1 Aerodynamic Drag

The equation for aerodynamic drag is given below:

$$F_D = \frac{1}{2} \rho C_d A v^2 \quad (13)$$

Where F_d is the aerodynamic drag, ρ is the air density, C_d is the drag coefficient of the vehicle, A is frontal area i.e. area of the vehicle normal to airflow. Lastly, v is the vehicle speed.

6.2 Rolling Resistance

Tyre rolling resistance which is a method to approximate the resistance that occurs due to the surface and tyre contact.

was also approximated using the formula:

$$F_R = C_{rr}mg \quad (14)$$

Where C_{rr} is the rolling resistance coefficient, m is the vehicle mass and g is the gravitational acceleration

As mentioned above, the data used in this project was synthetic and continuing with that approach further data was made and the values chosen for each term are listed in the table

Variable	Value
Air Density (ρ)	1.225 kg/m^3
Drag Coefficient (C_d)	0.32
Frontal Area (A)	2.2 m^2
Mass	1500 kg
Rolling Resistance Coefficient (C_{rr})	0.015
Gravitational Acceleration (g)	9.81 m/s^2

Table 4: Parameter Values to Determine Resistive Forces

Given the information listed in the table, its evident that for this model, the rolling resistance was modelled to be constant and a value of approximately 221 N was calculated.

The graph below shows how the resistance force(s) vary with vehicle speed. Its evident from the graph (and from the equation) that the aerodynamic drag, F_d is proportional to vehicle speed squared, v^2 . That relationship explains the quadratic shape of the aerodynamic drag (blue curve) as well as the total resistance, $F_{res} = F_D + F_R$ (orange curve).

The offset between the aforementioned curves is explained by the rolling resistance which was assumed to have be constant and calculated to have a value of approximately 221 N (dotted blue line). Additionally, the larger resistance (at lower speeds) is attributed to the rolling resistance however at larger speeds where $v \approx 84$ km/h the aerodynamic drag begins to become the larger source of resistance to the forward motion of the vehicle.

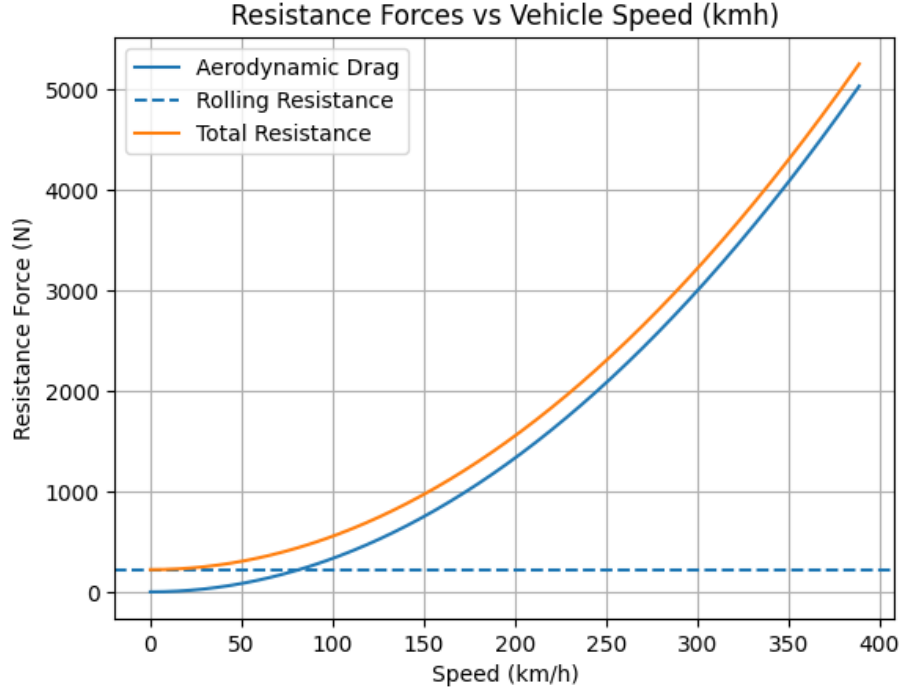


Figure 6: Resistance Forces vs Vehicle Speed

7 Longitudinal Dynamics and Time Simulation

7.1 The Net Force

The previous section was dedicated to determine the resistive forces and the section before determined the tractive force pushing the vehicle forward. Now the net force can be calculated using the forces from the previous section.

$$F_{net} = F_t - F_D - F_R \quad (15)$$

Where F_t is the tractive force, F_D is the aerodynamic drag force, and F_R is the rolling resistance.

Using the net force, F_{net} the acceleration at a point i.e. vehicle speed can be obtained using Newtons second law: $F_{net} = ma$

Rearranging for acceleration, a the formula becomes:

$$a = \frac{F_{net}}{m} \quad (16)$$

As previously mentioned the acceleration of the vehicle is not constant throughout and forces (tractive force and aerodynamic drag) vary with the vehicle speed.

7.2 The Simulation in Time

A time domain simulation was developed to determine the change in vehicle speed and position as the vehicle accelerated in a straight line.

At each time step (of 0.01 s):

1. Determined current vehicle speed
2. Selected the correct gear
3. Determined tractive force from gear curve
4. Determined Aerodynamic Drag and Rolling Resistance
5. Determined the Net Force
6. Update the Vehicle Velocity and position

The velocity was updated using the forward time step equation:

$$v_{n+1} = v_n + a_n \Delta t \quad (17)$$

while the position was updated using:

$$x_{n+1} = x_n + v_n \Delta t \quad (18)$$

Δt is the simulation time step.

At each time step, the net force, acceleration, speed, position and gear were stored in an array to allow for analysis post simulation.

7.3 Approximating the Launch

The engine torque begins at 1000 RPM whereas the simulation begins at rest. In a real vehicle the clutch would allow the engine to rotate while the vehicle is stationary and gradually transfer the torque from the engine to the wheel during the launch. However, this project did not model the clutch behaviour the tractive force at 1000 RPM in gear 1 was assumed to remain constant between 0 *km/h* and the minimum speed of the vehicle represented by the first gear curve. Once this minimum speed was reached, the normal tractive force model was used.

7.4 Simulated Acceleration Performance

The longitudinal simulation for this synthetic vehicle predicted a 0-100 *km/h* time of around 5.025s.

The graph below (7) shows how acceleration initially increases into the higher torque region its operating range. As the vehicle speed increases, the resistive forces (namely aerodynamic drag) and the reduction in engine torque at higher RPM causes acceleration to decrease. 7 also shows acceleration around the 4s

and approximately 82 km/h mark, which is when the gear shift (from gear 1 to gear 2 occurred), which can be seen much clearer in 9 below.

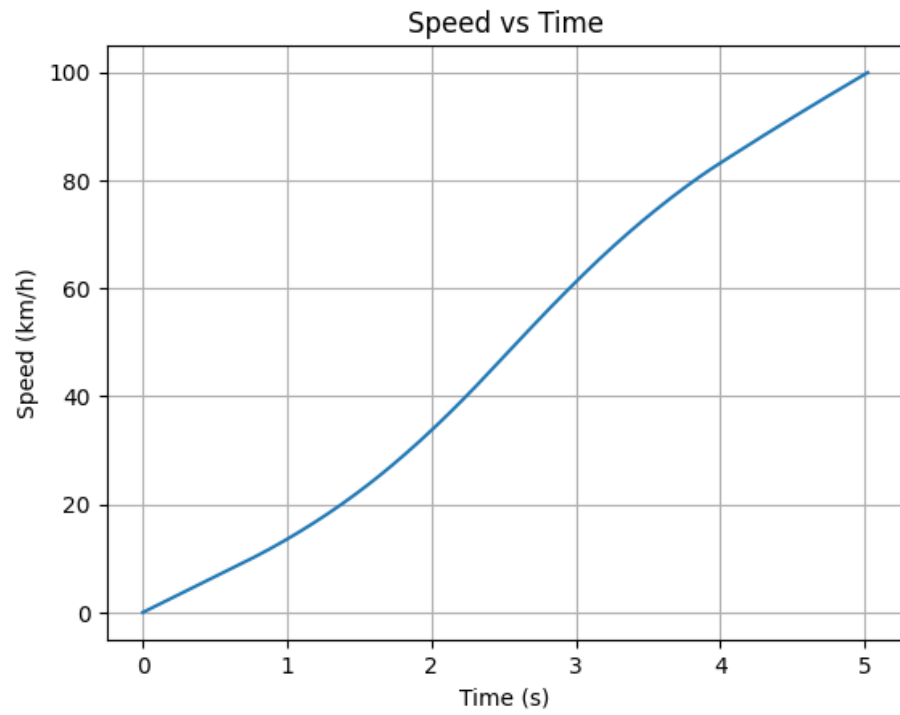


Figure 7: 0-100kmh time

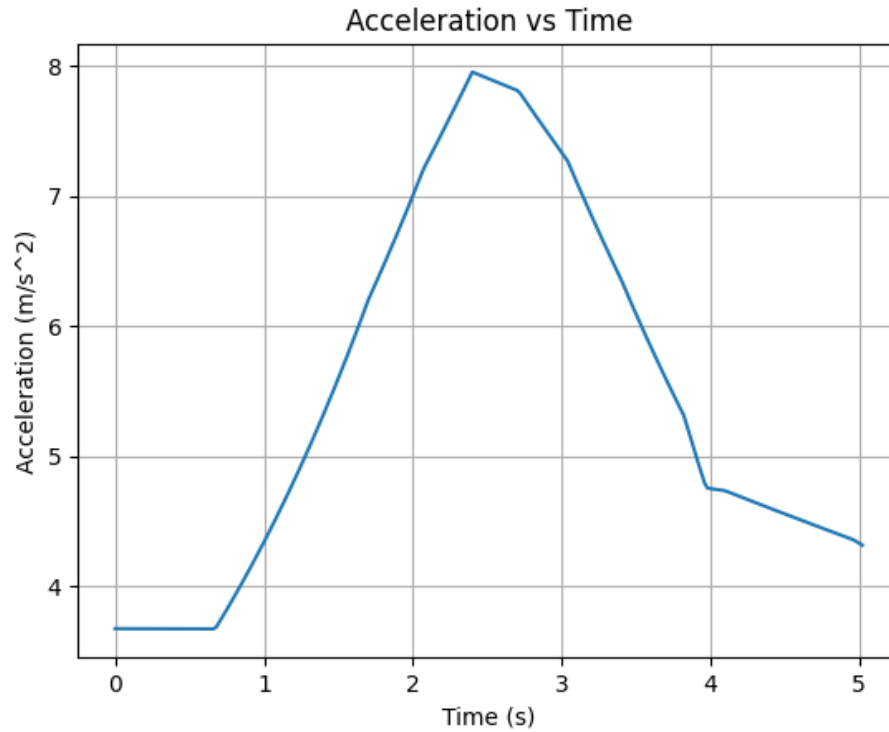


Figure 8: Acceleration vs Time

Figure 8 above is the acceleration time graph during the 0-100 km/h acceleration period of approximately 5 seconds. During the initial launch, acceleration remains constant because the tractive force that corresponds to 1000 RPM in gear 1 remains constant. Once the vehicle reaches the minimum speed, the increase in engine torque causes acceleration to increase. After the engine passes the peak torque region, the decreasing torque with increasing aerodynamic drag causes acceleration to decrease. At approximately the 4s point in figure ?? there is a slight 'kink' and acceleration is no longer decreasing at the same rate as before. This 'kink' is the gear change (gear 1 to gear 2)

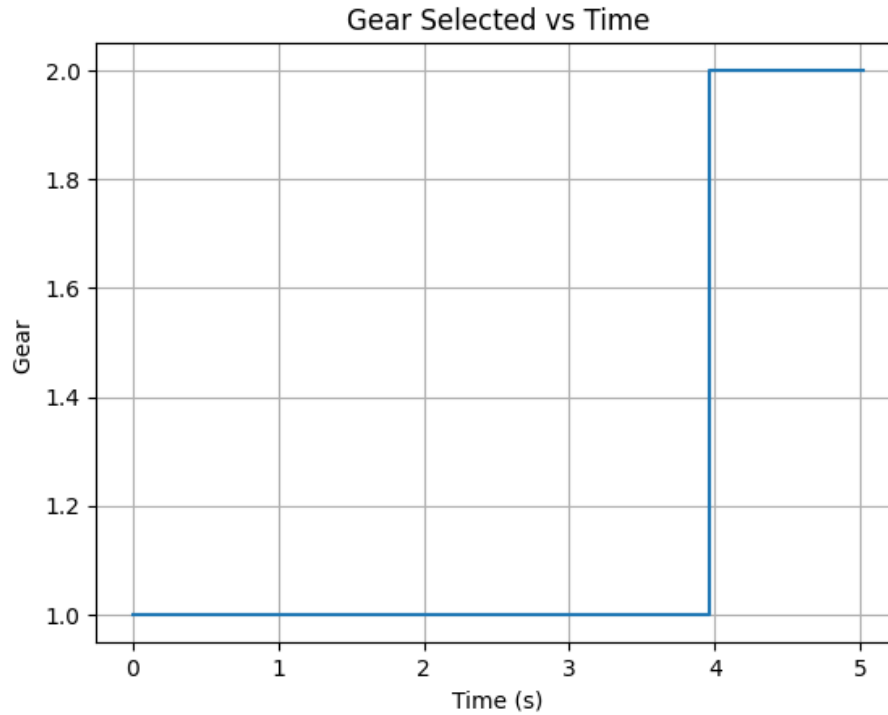


Figure 9: Gear 1 to 2

The figure above illustrates the gear change against time and further demonstrates the gear change from gear 1 to gear 2 around 4s point in time.

7.5 Force Behaviour

The individual and net forces were recorded throughout the simulation for further analysis, to see how they vary during the 0-100 *km/h* sprint. Figure 10 below shows the net and tractive force during the 5s. From a first glance it bears a striking resemblance to the acceleration vs time graph in figure ?? and this can be explained by considering the nature of the forces acting on the vehicle and Newton's second law. Vehicle acceleration is proportional to the net force applied onto the vehicle.

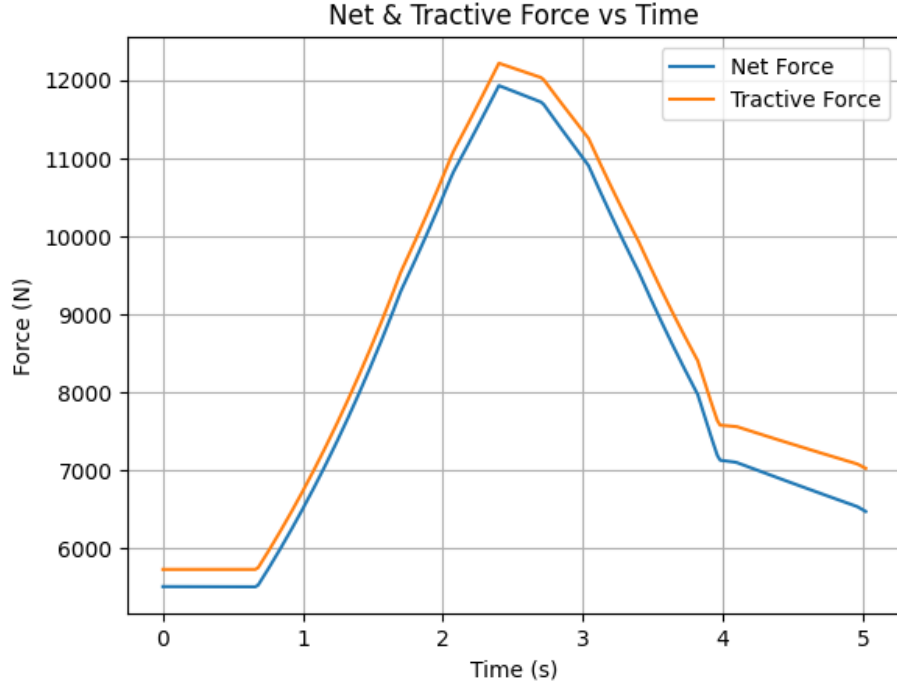


Figure 10: Net and Tractive Force

7.6 Influence of timestep on predictions

A small time step sensitivity study was also carried to determine if the simulation result was significantly influenced due to the time step chose.

Timestep, $\Delta t(s)$	Predicted 0-100 <i>km/h</i> Time <i>s</i>
0.020	5.040
0.010	5.030
0.005	5.025

Table 5: Timestep and Predictions

As shown above in 5 reducing the timestep from 0.02 to 0.01 resulted in a small change of 0.015s and a further decrease in the timestep to 0.005 resulted in no significant change. Hence the simulation was considered to be adequately converged at a timestep of 0.01s.

Shift Method	0-200 <i>km/h</i> Time(s)
Redline	18.9
Tractive Force	17.12

Table 6: Comparison of shifting methods for the 0-200 *km/h* event

8 Shift Strategy Optimisation and Performance Comparison

8.1 Baseline and Optimised Shifting Methods

For the evaluation, two approaches were considered using the same vehicle parameters and resistance models. The baseline strategy would stay in the same gear until the maximum RPM of 10,000 was reached i.e. until the engine had redlined.

The optimised strategy used the tractive force cross over points. An up-shift occurred when the tractive force available in the next gear was greater than the tractive force in the current gear. Since the rolling resistance and aerodynamic drag do not change due to the gears themselves, at a given vehicle speed, maximising the overall net force would therefore maximise vehicle acceleration. A simple function was developed that would take in different shift speed schedules.

8.2 From 0-200 *km/h*

A 0-200 *km/h* acceleration event was selected as a comparison method as it included multiple gear changes as opposed to the single gear change described in section above. As a result, this event provided a more representative assessment of the cumulative effect of shift optimisation.

From table 6 above it can be determine that the optimised strategy saved approximately 1.8s, which is equal to a reduction of approximately 9.4%

Figure 11 below shows the trajectory of the vehicle under the different strategies. The speeds initially follow a very similar path, because both methods use the same first gear model. As the vehicle progresses through the gears the curves begin to separate because the redline methodology stays in the current gear until it redlines whereas the optimised strategy will up-shift when the next gear provides a greater tractive force. As a result, the optimised strategy causes the vehicle to maintain a higher acceleration across a larger portion of the event thereby reaches the 200 *km/h* mark earlier.

0-200 km/h Acceleration Shift Strategy Comparison: Redline vs Optimised

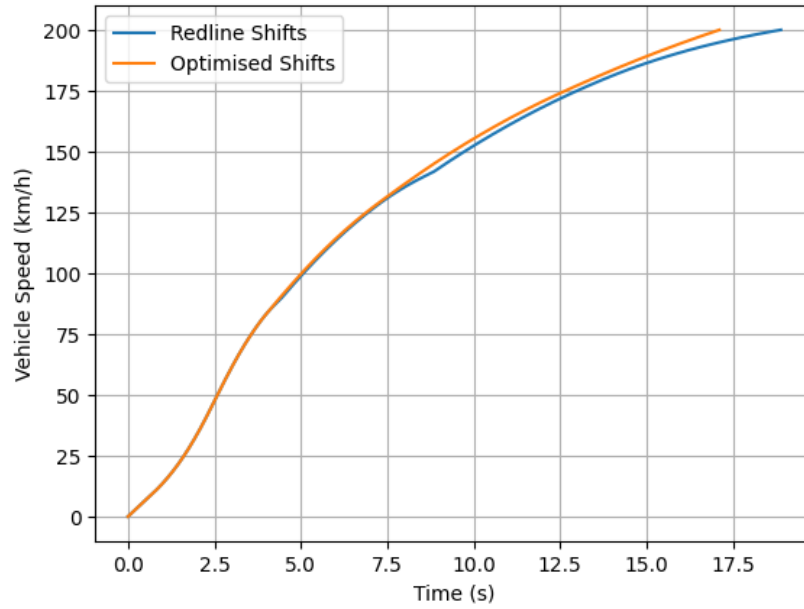


Figure 11: Redline vs Optimised

8.3 Acceleration and Gear Selection

Acceleration Comparison between Shift Strategies: Redline and Optimised

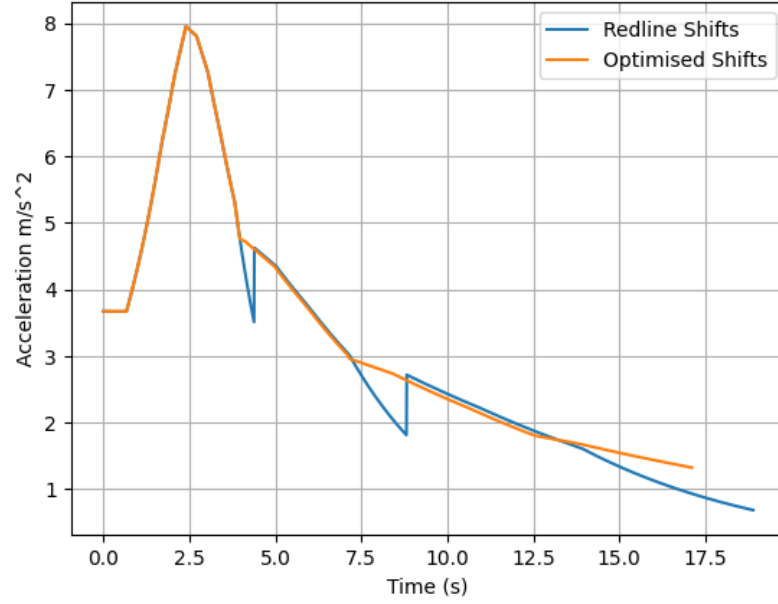


Figure 12: Acceleration Comparison between Shift Strategies

The acceleration comparison shows how under the redline strategy acceleration falls significantly towards the higher end of each gear as engine torque decreases at high engine speeds. The optimised strategy shifts much earlier (especially in lower gears) at the calculated tractive force cross-over points. Although the up-shift reduces the engine speed the higher torque in the next gear results in greater tractive force and for a vehicle with constant mass:

$$a_{i+1} > a_i \quad (19)$$

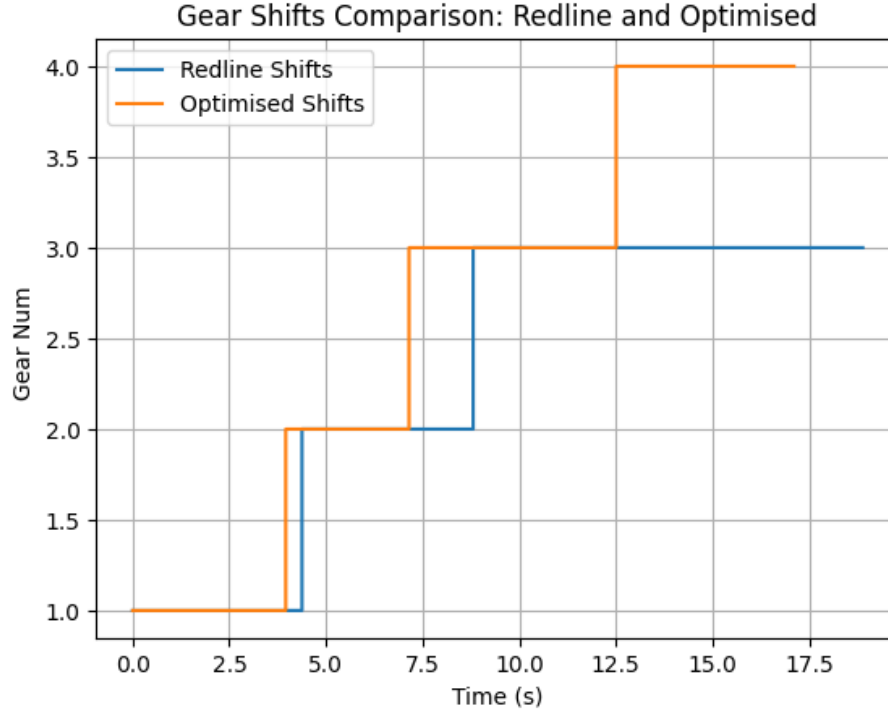


Figure 13: Gear Shifts Comparison: Redline and Optimised

The gear selections graph above further demonstrates that the optimised strategy resulted in earlier shifts relative to the redline strategy, allowing the engine to remain within the favourable high torque regions, in terms of wheel tractive force..

8.4 High Speeds

After the 0-200 km/h event another higher speed event was simulated. For this event the target speed was set to 350 km/h to investigate the high speed behaviour of both strategies. Although the gearing system technically allows for greater vehicle speeds, the simulation (even when extended to 1000s) did not reach speeds higher than 270 km/h . The optimised strategy had final top speed of 267.22 km/h and reached gear 6 whereas the redline strategy reached a top speed of 249.77 km/h and reached gear 4.

At these points the net longitudinal forces approached 0 Newtons and therefore the acceleration had approached at 0 m/s^2 as well. This demonstrates that the tractive force was balanced by the resistive aerodynamic drag and rolling resistance forces. This is shown below in figure 14

0-350 km/h Acceleration Shift Strategy Comparison: Redline vs Optimised

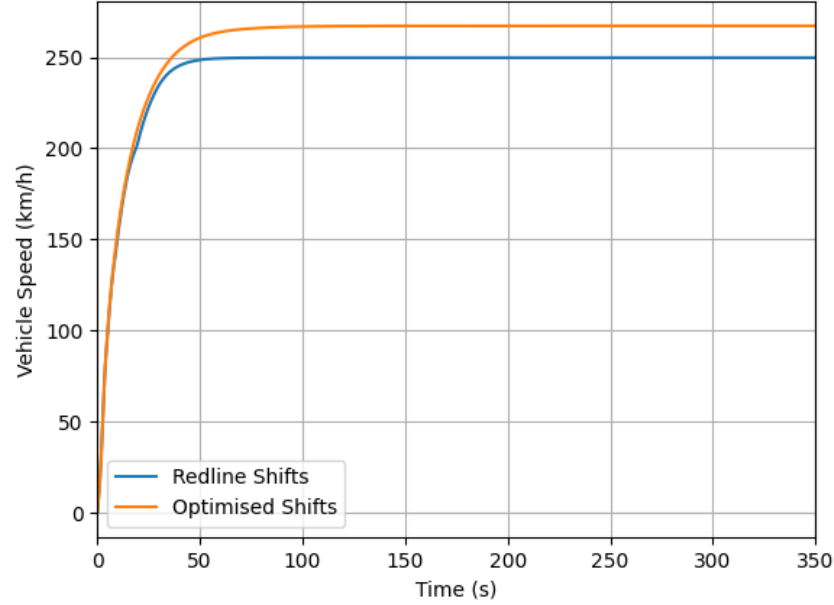


Figure 14: High Speed Simulation with a 350 km/h Target: Redline vs Optimised

Another difference between the two strategies was that the final speed of the redline strategy was achieved in gear 4 whereas the optimised strategy had climbed up to gear 6. This suggests that the redline strategy was unable to reach 10,000 RPM at a speed of nearly 250 *km/h* to be able shift into the next gear. However, the optimised strategy up-shifted to gear 6 and was able to reach a higher top speed.

8.5 Outcome

The optimised strategy produced two valuable outcomes for this study. The primary test of 0-200 *km/h* showed the optimised strategy saving nearly 1.8s which represents a 9.4% reduction relative to the redline strategy.

The secondary high speed investigation demonstrated that the optimised strategy achieved a higher top speed at around 267 *km/h* relative to redline shifting strategy achieving a top speed of approximately 250 *km/h*.

These results demonstrate how a first principles model can be used alongside simple optimisation to improve drivetrain decisions and its effects can be quantified for analysis.

9 Concluding Remarks, Study Limitations and Future Works

This study demonstrated from first principles the development of a synthetic engine model, tractive force that drive the vehicle as well as the resistive forces that would hinder the vehicle longitudinal motion. It demonstrated the consequences of utilising a force based methodology to optimise shifting and achieve both higher vehicle speeds and accelerations.

However, this study is not without limitations. The model contained multiple simplifying assumptions. Engine and vehicle parameters were synthetic, drivetrain efficiency and vehicle mass were assumed to remain constant and rolling resistance was also represented with a constant coefficient. Tyre traction limits, wheel slip and downforce were also some factors that were not included. Vehicle launch was represented using a constant low speed tractive force approximation instead of a clutch model. Gear changes were also assumed to be instantaneous with no torque transmission being interrupted. These assumptions allowed the model to focus on the relationship between engine characteristics, gearing and shift strategy but limit direct comparison with a real vehicle.

Future works would include modelling the engine with real manufacturer data for a given vehicle, accurate vehicle parameters such as mass and drag coefficient as well as further research into the rolling resistance coefficient and how it influences the total resistive forces instead of assuming it to be a constant factor.